Chengxi Yang

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Education

Shanghai Jiao Tong University Shanghai, China
Bachelor of Computer Science *Sep. 2022 - Jun. 2026 (expected)*

- A member of ACM Honors Class, an elite CS program in SJTU
- **GPA**: **92.7**/100, Ranking: 1/30, 22 A+ courses
- Selected Courses: Mathematical Logic: 100/100, Program Verification: 99/100, Model Checking: 100/100, Compiler Design and Implementation: 97/100, Cryptography: 100/100, Reinforcement Learning: 96/100

Research Interest

My research interest mainly lies in **programming languages** and **formal verification**, aiming to develop elegant theories to solve real-world problems. I'm also broadly interested in other areas related to PL, such as compiler design and AI4FTP (AI for Formal Theorem Proving).

Research Experience

University of Wisconsin-Madison Madison, WI, USA
Research Intern at madPL, co-advised by Prof. Thomas Reps and Prof. Tej Chajed *Jun. 2025 - present*

A Sound and Practical Foundation for Incremental Recursive Stream Computation (Ongoing)

As the project leader, I identified the "Fixedpoint Checking" (FPC) problem in incremental recursive stream computation models (e.g., DBSP), showing that naive termination checks are unsound while the exact specification is undecidable. I developed a sound, decidable and practical approximation (IntFP) to resolve this, enabling a **total correctness** Hoare logic and a **resource analysis** framework. The theory is formalized in **Lean**, with a paper in preparation for POPL/OOPSLA.

Shanghai Jiao Tong University Shanghai, China
Research Assistant, advised by Prof. Qinxiang Cao *Jun. 2024 - Jun. 2025*

Formalizing Temporal Reasoning Using the Trace Method (Ongoing)

I proposed a "**trace method**" to formalize temporal reasoning about program history, which is hard to capture in standard Hoare logic. I introduced novel proof rules for recursions and loops that faithfully reflect natural reasoning patterns. We have used this to formalize algorithms like DFS and Tarjan's bridge-finding algorithm in **Rocq**.

A Formal Framework for Naturally Specifying and Verifying Algorithms

I built a formal framework in Rocq, including a Hoare logic and tactics, for more natural algorithm verification. I proposed a novel **two-stage proof approach** that separates high-level logical reasoning from mechanical proof composition, better mirroring the logical structure of natural proofs. The paper is published in TASE 2025.

Contributing to the C Verification Tool QCP

I designed some core features for the QCP C verification tool, including "**multi-branch**", which enable fine-grained control over branches in symbolic execution. I also designed, implemented, and now maintain the tool's **VS Code extension** to improve usability, and I have contributed verifications of several examples to its backend Rocq library.

Publications and Preprints

A Formal Framework for Naturally Specifying and Verifying Sequential Algorithms
Chengxi Yang*, *Shushu Wu**, *Qinxiang Cao*
In Theoretical Aspects of Software Engineering (TASE) 2025. [arxiv] [code]

QCP: A Practical Separation Logic-based C Program Verification Tool
Xiwei Wu, Yueyang Feng, Xiaoyang Lu*, Tianchuan Lin, Kan Liu, Zhiyi Wang, Shushu Wu, Lihan Xie,* ***Chengxi Yang****, Hongyi Zhong, Naijun Zhan, Zhenjiang Hu, Qinxiang Cao*

Under submission to FM 2026. [arxiv]

Selected Projects

- **Mx* Compiler**: A Compiler from Mx* language (a C & Java like language) to RISC-V RV32I Assembly implemented in Java, featuring linear scan register allocation, SSA transformation and various optimization.
- **LTL Model Checker**: A Linear Temporal Logic (LTL) model checker implemented in Rust. Given a transition system (TS) and an LTL formula, it can check whether the TS or its arbitrary state satisfies the formula.
- **RISC-V CPU**: A Tomasulo-based RISC-V RV32I CPU designed in Verilog RTL, featuring instruction cache and branch predictor. It can be run on FPGA board.
- **Acore**: A toy RISC-V kernel implemented in Rust that can be run on QEMU. It features remote-procedure-call mechanism, virtual memory and preemptive multi-process scheduling.
- **Resource-Constrained MOBA Agent**: We optimized PPO-based agents for Honor of Kings under limited computational resources by making reward dynamic and denser, integrating Phasic Policy Gradient (PPG) and mixed opponent selection.

Honors & Awards

Scholarship

- **OPPO "Benfen" Scholarship** (first-year winner, about 30 winners in SJTU), 2025
- **Fan Xuji Scholarship** (about 15 winners each year in SJTU), 2024
- **Zhiyuan Honorary Scholarship** (awarded to top 10% students in SJTU annually), 2022-2025

Teaching

- **TA for Programming(C++)**. I gave a lecture on introduction to computer systems, designed a new course project and a problem in the final exam.
- **TA for Principles and Practice of Computer Algorithms**. I designed a new lab project on formal verification in Coq.
- **TA for Mathematical Logic**. I answered students' questions, graded their homework and wrote sample solutions for them.
- **TA for Program Verification**. I maintained the course git repository and participated in the design of the final project.

Skills

Natural Languages: Chinese (native), English (fluent, TOEFL 108).
Programming Languages: Rocq (Coq), Lean, Rust, C/C++, Java, Python, Verilog, Go.